NFC Institute of Engineering & Technology, Multan

Department of Computer Science

Ref. No.IET/Comp.Sc/4411 Dated: March 6, 2026

NOTIFICATION

Subject: Mandatory Computer-Based Test for Final Year Computing Students

All students are hereby informed that the **Higher Education Commission in** collaboration with the **Ministry of Information Technology & Telecommunication, Pakistan Software Export Board, Pakistan Software Houses Association, and National Computing Education Accreditation Council**, has introduced a **mandatory Computer-Based Test** for final year computing students. The **Virtual University of Pakistan (VU)** has been designated as the official testing body responsible for conducting the test.

All students enrolled in the **8th semester** are required to appear in this test as it carries **1 credit hour** and is **mandatory for degree completion**.

Eligibility:

Students currently enrolled in the 8th semester of the Computer Science and Software Engineering programs at HEC-recognized universities are eligible to appear in the test.

Test Schedule:

The test is scheduled to be conducted on April 04–05, 2026.

Test Format:

- Computer-Based Test
- **100 Multiple Choice Questions (MCQs)**
- Conducted by **Virtual University of Pakistan** at designated test centers in each city

Competency Areas:

The test will assess **industry-endorsed competency areas** relevant to computing disciplines. Detailed competency areas are available in **Annexure-A**.

All eligible students are advised to **ensure their participation** in the test as per the scheduled dates as it is mandatory for degree completion.

Dr. Naeem Aslam
Head, Department of Computer Science

Annexure-A

Areas of Competencies for National Skill Competency Test of IT Graduates

Arears of Competencies	Weightage %	Topics
Computer Networks and Cloud Computing	10%	1 - Data Communication
		2 - Computer Networks
		3 - Data Link Layer
		4 - Network Layer
		5 - Transport Layer
		6 - Application Layer
		7 - Wireless Networks
		8 - Cloud Computing
		9 - Network Security (Networks Perspective)
		10 - Next Generation Networks
Programming (C++/Java/Python)	10%	1. Programming Fundamentals
		2. Data Types & Variables
		3. Operators & Expressions
		4. Control Structures
		5. Functions / Methods,
		6. Input / Output Handling,
		7. Strings & Text Processing
		8. Arrays & Collections
		9. Object-Oriented Programming (OOP)
		10. Memory Management Concepts
		11. Exception & Error Handling
		12. Modules, Packages & Libraries
		13. Advanced Programming Concepts
		14. Concurrency & Parallelism (Introductory)
		15. Debugging, Testing & Optimization
		16. Software Development Practices
Data Structures & Algorithms	10%	1 - Foundations of Data Structure and Algorithms
		2 - Linear Data Structures
		3 - Non-Linear Data Structures
		4 - Searching Algorithms
		5 - Sorting Algorithms
		6 - Hashing
		7 - Tree Algorithms
		8 - Graph Algorithms
		9 - Algorithm Design Techniques
		10 - Advanced Data Structures
		11 - String Algorithms
		12 - Complexity & Optimization
Operating Systems	5%	1. Introduction to Operating Systems
		2. Operating System Structures
		3. Process Management
		4. CPU Scheduling
		5. Thread Management

		6. Concurrency & Synchronization
		7. Deadlocks
		8. Memory Management
		9. File System Management
		10. Secondary Storage Management
		11. Input / Output Systems
		12. Protection & Security
Software Engineering	10%	1. Introduction to Software Engineering
		2. Software Process Models
		3. Agile Software Development
		4. Software Requirements Engineering
		5. Software Project Management
		6. Software Design
		7. Software Architecture
		8. User Interface Design
		9. Software Implementation & Coding
		10. Software Testing
		11. Software Maintenance & Evolution
		12. Software Quality Assurance
		13. Software Metrics & Measurement
		14. Software Configuration Management
		15. Software Risk Management
		16. Software Security Engineering
Web Development	10%	1. Introduction to Web Development
		2. Web Architecture & Protocols
		3. HTML Fundamentals
		4. CSS Fundamentals
		5. Advanced CSS & Responsive Design
		6. JavaScript Fundamentals
		7. Advanced JavaScript
		8. Frontend Frameworks & Libraries
		9. Backend Development Fundamentals
		10. Server-Side Programming
		11. Databases for Web Applications
		12. Web Security
		13. Web Performance & Optimization
		14. Web Testing & Debugging
		15. Deployment & Hosting
		16. Web APIs & Integration
		17. Modern Web Development Practices
AI / Machine Learning and Data Analytics	10%	1. Introduction to AI, ML & Data Analytics
		2. Mathematical Foundations
		3. Python for AI & Data Analytics
		4. Data Collection & Pre-processing
		5. Exploratory Data Analysis (EDA)
		6. Supervised Learning

		7. Ensemble Learning
		8. Unsupervised Learning
		9. Model Evaluation & Validation
		10. Feature Engineering & Selection
		11. Deep Learning Fundamentals
		12. Advanced Deep Learning
		13. Natural Language Processing (NLP)
		14. Computer Vision
		15. Big Data Analytics (Introductory)
		16. Model Deployment & MLOps Basics
		17. AI Ethics, Security & Privacy
Cyber Security	5%	1. Introduction to Cyber Security
		2. Security Fundamentals & Principles
		3. Cryptography Basics
		4. Network Security
		5. Operating System Security
		6. Web Application Security
		7. Malware & Attack Techniques
		8. Authentication & Access Control
		9. Secure Software Development
		10. Wireless & Mobile Security
		11. Cloud & Virtualization Security
		12. Digital Forensics
		13. Incident Response & Management
		14. Security Monitoring & Auditing
		15. Cyber Laws & Ethics
		16. Emerging Trends in Cyber Security
Databases	10%	1. Introduction to Database Systems
		2. Database System Architecture
		3. Data Models
		4. Relational Database Concepts
		5. Relational Algebra & Calculus
		6. Structured Query Language (SQL)
		7. Advanced SQL
		8. Database Design & Normalization
		9. Transaction Management
		10. Concurrency Control
		11. Recovery Management
		12. Indexing & File Organization
		13. Query Processing & Optimization
		14. Database Security
		15. Distributed Databases
		16. NoSQL & Modern Databases
		17. Data Warehousing & Data Mining (Introductory)
	20%	1. Introduction to Problem Solving
		2. Problem Understanding & Analysis

Problem Solving And Anaytical Skills		3. Logical Reasoning Fundamentals
		4. Algorithms & Flow Control
		5. Data Representation & Abstraction
		6. Pattern Recognition & Generalization
		7. Mathematical & Quantitative Reasoning
		8. Algorithmic Thinking
		9. Critical Thinking & Decision Making
		10. Debugging & Error Analysis
		11. Complexity & Efficiency Awareness
		12. Problem Solving Using Programming
		13. Data-Driven Problem Solving
		14. Creative & Innovative Thinking
		15. Real-World Problem Solving
		16. Communication & Documentation of Solutions